Prabhavit Gupta

Frontier Agents Engineer | Applied AI, Multi-Agent Systems & Production Evaluation Harnesses

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Executive Summary

- **Frontier AI Agents & Applied Reasoning Architectures:** Applied software engineer with 3 years commercial software experience architecting production AI agent systems, multi-agent coordination loops, and grounded tool-calling harnesses for complex customer workflows.
- **Rigorous Evaluation, Benchmarking & AI Quality:** Hands-on practitioner designing quantitative evaluation frameworks using offline benchmarks, golden datasets, prompt regression test suites, and LLM-as-a-judge scoring (Langfuse, DeepEval) to systematically prevent quality drift.
- **Production AI Engineering & Full-Stack Delivery:** Proven engineering ability combining asynchronous Python runtimes (FastAPI, AsyncIO), TypeScript/Next.js interfaces, Model Context Protocol (MCP) servers, and enterprise data integrations into reliable software systems.

Technical Skills & Core Competencies

- **Frontier Agents & Reasoning Paradigms:** Multi-Agent Orchestration, Plan-and-Execute, ReAct Frameworks, Autonomous Tool Calling, Model Context Protocol (MCP), Context Management, Human-in-the-Loop Safeguards, Dynamic Prompting.
- **Evaluation & AI Observability:** Quantitative Eval Frameworks (Langfuse, DeepEval), Golden Datasets, Offline Benchmarks, Prompt Regression Suites, LLM-as-a-Judge, Distributed Tracing, Hallucination Reduction, Guardrails.
- **Applied AI & Backend Engineering:** Python (AsyncIO, FastAPI, PyTest), TypeScript, Next.js, Node.js, Structured Outputs (Pydantic), RESTful APIs, GraphQL, PostgreSQL, SQLite, Redis, Vector Databases / RAG.
- **Systems & Production Infrastructure:** Docker, Kubernetes, Linux CLI, GitOps / GitHub Actions CI/CD, Enterprise Data Ingestion, Security Best Practices, Technical Scoping & Client Advisory.

Professional Experience

Forward Deployed AI Engineer & Applied AI Consultant • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Designed and deployed production-grade AI agent systems for enterprise clients, translating complex business processes into multi-step agent planning and tool execution pipelines.
- Developed custom Model Context Protocol (MCP) servers and tool-calling harnesses connecting reasoning models to internal customer databases, APIs, and file repositories.
- Built automated evaluation pipelines in Langfuse, measuring agent planning trajectories, tool call precision, and semantic correctness against curated golden benchmark suites.
- Implemented deterministic validation layers and human-in-the-loop review triggers for high-impact actions, ensuring safety, reliability, and regulatory compliance.
- Collaborated with customer technical leads to rapidly prototype new agent capabilities, evaluate emerging frontier models, and promote validated prototypes into production.

Lead Developer & Product Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Led core platform development and data integration engineering, scaling the system to support 4,500+ active enterprise users with high reliability.
- Re-architected backend microservices onto Next.js and high-concurrency API backends, establishing zero-downtime deployment pipelines and strict data validation layers.
- Built asynchronous data ingestion pipelines collecting and transforming operational inventory from external APIs with automated fault isolation and retry logic.
- Engineered automated incident triage workflows and webhook integrations (OfficeRnD, Nexudus) ensuring real-time state synchronization across distributed booking resources.
- Integrated automated conversational support workflows connected to live operational databases, reducing manual triage workload while maintaining high resolution accuracy.

Full-Stack Solutions Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built custom API integrations and automated backend microservices in Python and TypeScript, bridging client platforms with internal enterprise data warehouses.
- Implemented containerized automated testing suites within Docker CI/CD pipelines, establishing reliable integration baselines.

- Partnered directly with technical and business stakeholders to translate complex business processes into robust software specifications.

Key Engineering Projects

- **Production AI Agent Evaluation & Tool Harness (TypeScript + Python + MCP + Langfuse):** Architected an execution harness enabling LLMs to safely invoke external tools over MCP, validating arguments against strict schemas with automated regression tracking and LLM-as-a-judge scoring.
- **Universal Multi-Source Data Ingestion Engine (Python + Playwright + REST APIs):** Designed an automated data harvesting and enrichment pipeline parsing 900+ structured and semi-structured external sources with schema validation and error isolation.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to international relocation to San Francisco or New York with visa sponsorship; available immediately.